

2(a)	any one from: <table><tr><td>tissue</td><td>reason ;</td></tr><tr><td>xylem</td><td rowspan="3">transports water</td></tr><tr><td>epidermis of root</td></tr><tr><td>cortex of root / parenchyma in root / endodermis</td></tr><tr><td>epidermis of leaf</td><td rowspan="2">(too much) water loss</td></tr><tr><td>mesophyll (of leaf)</td></tr></table>	tissue	reason ;	xylem	transports water	epidermis of root	cortex of root / parenchyma in root / endodermis	epidermis of leaf	(too much) water loss	mesophyll (of leaf)	1
tissue	reason ;										
xylem	transports water										
epidermis of root											
cortex of root / parenchyma in root / endodermis											
epidermis of leaf	(too much) water loss										
mesophyll (of leaf)											
2(b)	it is not a 3:1 ratio (expected) ; reason (1 of:) due to interactions between genes ; non resistant plants / certain allele combinations, have reduced chances of survival ;	2									

Question	Answer	Marks
2(c)(i)	<i>any 3 of:</i> 1 probes / known DNA sequences attached to plate / slide / AW ; 2 <i>ref. to</i> (probe) DNA is single stranded ; 3 <u>mRNA</u> from resistant and non-resistant plants are extracted ; 4 <u>mRNA</u> converted to <u>cDNA</u> ; 5 cDNA / single strand of DNA (from samples) labelled with different colour (fluorescent) dyes (for resistant and non-resistant plants) ; 6 cDNA / single strand of DNA (from sample), allowed to hybridise with probe / added to plates / added to probes ; 7 fluorescence identifies presence of genes ;	3
2(c)(ii)	3 : 1 ;	1
2(c)(iii)	67 / 66.6 ;	1
2(c)(iv)	at 24 hours, genes expressed more rapidly / more genes expressed, in resistant variety compared to non-resistant variety / ora or at 48 hours, less genes expressed, in resistant variety compared to non-resistant variety / ora ; number of genes expressed decreases in resistant variety as time increases ;	2